

Eventual Sign Coherence

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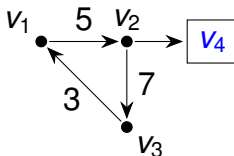
Algebraic and Enumerative Combinatorics Seminar,
University of Waterloo,
June 2026

joint work with
Amanda Burcroff
[arXiv:2605.12865](https://arxiv.org/abs/2605.12865)

Quivers

Definition

A **quiver** is a finite directed graph. Multiple arrows are allowed, but no loops nor oriented 2-cycles. The arrow multiplicities are called **weights**. Vertices are marked as **frozen** $\square$ or **mutable** $\bullet$.



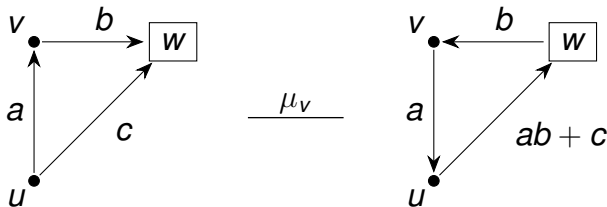
Arrows between frozens play no role in the combinatorics (or algebra), so they are generally omitted.

Quiver mutation

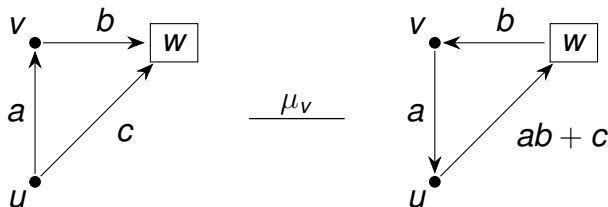
Definition

The quiver **mutation** $\mu_v(Q) = Q'$ is computed in three steps.

- If $u \xrightarrow{a} v \xrightarrow{b} w$ then add $a \cdot b$ arrows $u \longrightarrow w$.
- Reverse all arrows incident to v .
- Delete oriented 2-cycles, one by one.



Quiver mutation



Proposition

Mutation is an involution.

Definition

A sequence of mutations is called **reduced** if it never mutates at the same vertex consecutively.

Subquivers and mutation

Definition

A **subquiver** is a (full) induced subgraph. Denoted by $Q|_S$, where S is some set of vertices.

Proposition

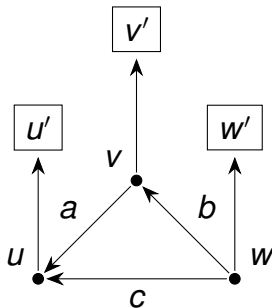
If v is a vertex in S then

$$\mu_v(Q)|_S = \mu_v(Q|_S).$$

Our pal, the Principal extension

Definition

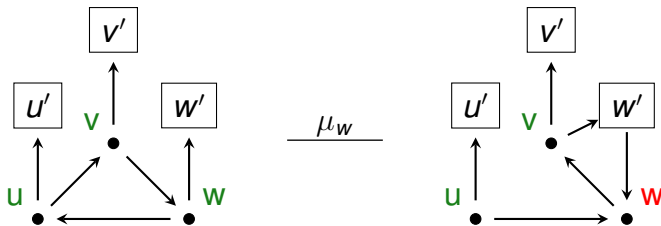
To construct the **principal extension** $\hat{Q}$ of a quiver Q , add an additional frozen vertex v' for each mutable vertex v , along with an arrow $v \rightarrow v'$.



Colouring instructions

Definition

A mutable vertex v is **red** (resp **green**) if all arrows between v and frozen vertices are oriented toward (resp. away) from v .



Colouring is possible

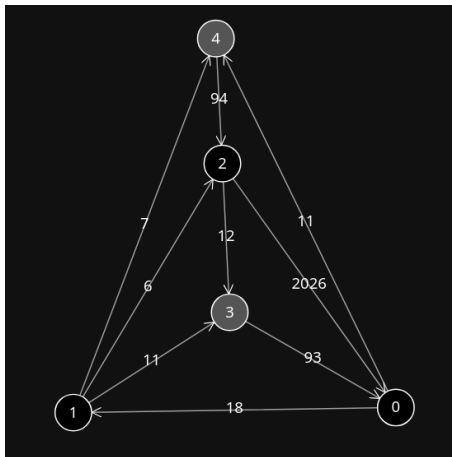
Definition

A mutable vertex v is **red** (resp **green**) if all arrows between v and frozen vertices are oriented toward (resp. away) from v .

Theorem [Derksen-Weyman-Zelevinsky 10]

Sign coherence: In every quiver mutation equivalent to $\hat{Q}$, every vertex is either red or green.

Sign coherence: inevitable?



Eventual sign coherence conjecture

Definition

A (possibly infinite) mutation sequence $M = v_1, v_2, \dots$ is **simple** for a quiver Q if the quivers $Q, \mu_{v_1}(Q), \mu_{v_2} \circ \mu_{v_1}(Q) \dots$ are pairwise distinct.

Conjecture [Gehktman-Nakanishi 22, Burcroff-N* 26]

If M is a simple infinite mutation sequence for a quiver Q , then

$$\mu_{v_j} \circ \dots \circ \mu_{v_1}(Q)$$

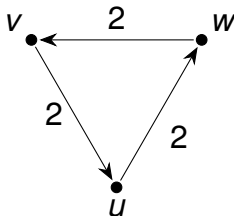
is sign coherent for all sufficiently large j .

If the conjecture holds for Q and $\mathbf{M}$, we say Q is **eventually sign coherent** on $\mathbf{M}$.

Previous results

Due to [Gehktman-Nakanishi 22], we know that the conjecture holds when:

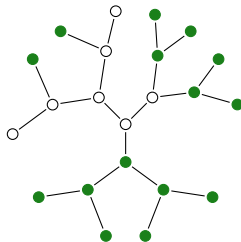
- Q has two mutable vertices,
- Q is the following quiver, and our sequence is $\mathbf{M} = \overline{uvw}$:



Almost always eventual sign coherence

Theorem [Burcroff-N 26]

Let Q be a quiver such that there are at least 3 arrows between some pair of mutable vertices. Let $\mathbf{M}$ be a random infinite reduced mutation sequence for Q . With probability 1, Q is eventually sign coherent on $\mathbf{M}$.



Forks

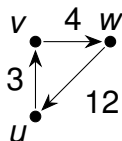
Definition

A quiver is **abundant** if there are at least 2 arrows between every pair of vertices.

Definition [Warkentin 14]

A **fork** F is an abundant quiver with a vertex r , called the *point of return*, such that:

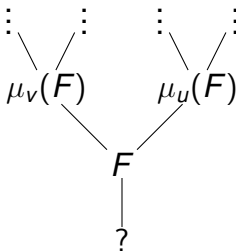
- F is not acyclic but F without r is;
- $i \xrightarrow{a} r \xrightarrow{b} j$ implies $j \xrightarrow{c} i$ with $c > \max(a, b)$.



Tree Lemma

Lemma [Warkentin 14]

If F is a fork and j is not the point of return, then $\mu_j(F)$ is a fork with point of return j .



Forks are always near

Theorem [Warkentin 14]

Suppose Q has at least 3 arrows between some pair of vertices. Then Q is at most n^3 mutations away from a fork, where n is the number of mutable vertices.

Forks stay coherent

Theorem [Ervin 24, Burcroff-N 26]

Let Q be a quiver with the mutable part of Q a fork and $\mathbf{M}$ be a reduced mutation sequence that does not start by mutating at the point of return. Under mild assumptions, if every vertex in Q is either red or green, then every vertex will be red or green in every quiver along $\mathbf{M}$.

Sign coherence emerges from forks

Definition

A quiver Q is an **ice fork** if the mutable part of Q is a fork, and whenever we ‘thaw’ one frozen vertex the mutable part remains a fork.

Theorem [Burcroff-N 26]

If Q is an ice fork, then any reduced mutation sequence which

- mutates at all vertices at least twice, and
- does not start with a mutation at the point of return

will result in a quiver where every vertex is red or green.

Almost always eventual sign coherence

Theorem [Burcroff-N 26]

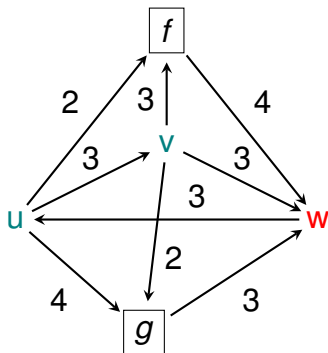
Let Q be a quiver such that there are at least 3 arrows between some pair of mutable vertices. Let $\mathbf{M}$ be a random infinite reduced mutation sequence for Q . With probability 1, Q is eventually sign coherent on $\mathbf{M}$.

Proof:

- 1 we have some probability of mutating to an ice fork,
- 2 (tree lemma) most mutations of ice forks result in an ice fork, and
- 3 mutating an ice fork enough times results in sign-coherence.

Non-principle, still coherent

Every vertex in every quiver in the mutation class of the following quiver is red or green [Burcroff-N 26].



More examples

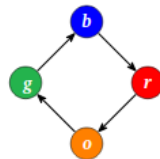
Theorem [Burcroff-N 26]

Suppose Q is a mutation-abundant quiver with 3 mutable vertices. Then Q is eventually sign coherent on any simple mutation sequence.

BROG



(Photo credit: vorpall reaper on Xenoblade Wiki)



Brog quivers

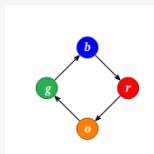
Definition

Two vertices u, v are called **complimentary** if for each frozen vertex f , $f \rightarrow u$ if and only if $f \leftarrow v$.

For example: a **red** and a **green** vertex are complimentary.

Definition

An abundant quiver with exactly 2 frozen vertices is called **brog** if its vertices can be colored blue, red, orange, and green so that: the blue and orange vertices are complimentary, and arrows are oriented as indicated below:



Brogerties

Proposition [Burcroff-N 26]

If Q is brog with 3 mutable vertices, then $\mu_j(Q)$ is either acyclic or brog.

(This holds in much greater generality.)

Proposition

Suppose Q is a quiver and $\mathbf{M}$ a reduced mutation sequence on Q such that every vertex is mutated at least twice, and every quiver we visit on the mutation sequence is brog. Then Q is eventually sign coherent on $\mathbf{M}$.

Thank you!

Ice embeddability

Definition

Say we have a quiver Q with 2 frozen vertices u, v . We say that $k \in \mathbb{Z}$ is **ice embeddable** if we can mutate Q so that $u \xrightarrow{k} v$.

Ice-embeddability is provably difficult (though embeddability is currently open!) [Soukup 23].

Theorem [Burcroff-N 26+]

Suppose Q has a finite forkless part when u, v are thawed. Then only finitely many integers are ice embeddable into Q .